SuffAI Command Center — GCP Redeploy Guide

Covers: AURA forms, 1P Confirm flow, Style Assistant picks computed at Extract & Rank, “Select all” filters, Geo and Admin updates. Run all commands from the repo root with `gcloud` authenticated.

What needs redeploying

Piece	Redeploy?	Why
`suffai-command-center`	**Yes**	All app and UI changes, plus the new `numpy` dependency
Cloud SQL migrations 005–010	**Yes, if not already applied**	Campaigns, ranking jobs, recommendations and Geo caches need them
`suffai-similarity`	**Yes, if `compare-pins` isn't live**	`similarity/compare.py` and `main.py` changed
Cloud Tasks queue	**Strongly recommended**	Ranking plus personalisation is too long to run safely in a Cloud Run background thread
`suffai-extraction`	No	Unchanged

Step 0 — Set variables and snapshot current settings

```
cd ~/Downloads/AI_CUP/AI_CUP_latest
export GOOGLE_CLOUD_PROJECT=YOUR_PROJECT_ID
export GCP_REGION=asia-south1
gcloud config set project "$GOOGLE_CLOUD_PROJECT"
gcloud config set run/region "$GCP_REGION"

# Save current env so nothing gets lost
for s in suffai-command-center suffai-similarity suffai-extraction; do
  gcloud run services describe $s --format=export > /tmp/$s.yaml 2>/dev/null || true
done

export EXTRACTION_URL=$(gcloud run services describe suffai-extraction --
format='value(status.url)')
export SIMILARITY_URL=$(gcloud run services describe suffai-similarity --
format='value(status.url)')
export CLOUD_SQL_INSTANCE="$GOOGLE_CLOUD_PROJECT:$GCP_REGION:suffai-ops"   # adjust instance
name
export DATABASE_URL='postgresql://USER:PASS@/DBNAME?host=/cloudsql/'"$CLOUD_SQL_INSTANCE"
```

> Copy your real DATABASE_URL from `/tmp/suffai-command-center.yaml` or from Secret Manager. Don't guess it.

Step 1 — Keep plaintext passwords out of the image (before building)

`Dockerfile.command_center` copies the whole `command_center_data/` folder into the image, and two files in it hold plaintext passwords. Move them outside the repo:

```
mkdir -p ~/suffai-secrets
mv command_center_data/auth/login_creds_plaintext.csv ~/suffai-secrets/
mv command_center_data/auth/brand_logins_summary.csv ~/suffai-secrets/
```

Create a `.dockerignore` at the repo root:

```
command_center_data/auth/login_creds_plaintext.csv
command_center_data/auth/brand_logins_summary.csv
.git
**/__pycache__
**/.DS_Store
```

Logins should come from BigQuery. Load the hashed credentials file (the script always writes to the `suffai_ops` dataset):

```
./command_center/scripts/load_login_creds_bq.sh "$GOOGLE_CLOUD_PROJECT"
```

Step 2 — Apply Cloud SQL migrations 005–010

Every migration is safe to re-run. Apply them in order (also 001–004 if they're not already in place).

```
cloud-sql-proxy "$CLOUD_SQL_INSTANCE" --port=5432 &
export PGPASSWORD='YOUR_DB_PASSWORD'
for f in 005_brand_sku_campaigns 006_campaign_1p_lifecycle 007_campaign_impact_metrics \
         008_campaign_journey 009_campaign_jobs_channels 010_geo_store_intel; do
  psql "host=127.0.0.1 port=5432 user=YOUR_DB_USER dbname=YOUR_DB_NAME" \
       -v ON_ERROR_STOP=1 -f db/migrations/$f.sql
done
kill %1
```

Check that the tables exist:

```
SELECT table_name FROM information_schema.tables
WHERE table_name IN ('brand_campaigns','brand_skus','campaign_jobs',
  'campaign_user_recommendations','campaign_impact_daily','geo_store_lookups');
```

> Ignore `deploy/01_prepare_similarity_db.sh` for this step — it only applies 003 and 004.

Step 3 — Redeploy `suffai-similarity` (only if `compare-pins` isn't live)

> This script replaces **every** setting on the service (`--set-env-vars`). Export `DATABASE_URL` and `CLOUD_SQL_INSTANCE` from Step 0 first.

```
./deploy/deploy_similarity.sh
```

Check it:

```
curl -s -X POST "$SIMILARITY_URL/v1/similarity/compare-pins" \
  -H "Authorization: Bearer $(gcloud auth print-identity-token)" \
  -H 'Content-Type: application/json' \
  -d '{"gcs_object_key":"demo/x.jpg","pin_codes":["560008"]}' | head -c 300
```

A 404 means the old version is still running. Any error about the image itself means the route exists.

Step 4 — Rebuild and redeploy `suffai-command-center`

Option A — local Docker (keeps existing settings via `--update-env-vars`)

```
./deploy/deploy_command_center.sh
```

Option B — Cloud Build (no local Docker)

```
gcloud builds submit --config=deploy/cloudbuild-command-center.yaml .
gcloud run deploy suffai-command-center \
  --image="$GCP_REGION-docker.pkg.dev/$GOOGLE_CLOUD_PROJECT/suffai/suffai-command-center:latest"
\
  --allow-unauthenticated --memory=1Gi --cpu=1 \
  --add-cloudsql-instances="$CLOUD_SQL_INSTANCE" \
  --update-env-
vars="USE_BQ=true,BQ_DATASET=suffai_ops,GOOGLE_CLOUD_PROJECT=$GOOGLE_CLOUD_PROJECT,GCS_IMAGE_BUC
KET=$GOOGLE_CLOUD_PROJECT-suffai-garment-
images,ENABLE_VERTEX=true,VERTEX_LOCATION=$GCP_REGION,VERTEX_CHAT_MODEL=gemini-2.5-
flash,EXTRACTION_URL=$EXTRACTION_URL,SIMILARITY_URL=$SIMILARITY_URL,DATABASE_URL=$DATABASE_URL"
```

Then switch logins to BigQuery only and set the Geo defaults:

```
gcloud run services update suffai-command-center \
  --update-env-vars="AUTH_SOURCE=bq,GEO_GROUNDING_LOCATION=global,GEO_CITY=Bengaluru"
```

Step 5 — Cloud Tasks for ranking jobs (one-time setup)

Without this, the job that ranks products and saves Style picks runs in a background thread that Cloud Run can freeze between requests.

```
PROJECT_NUMBER=$(gcloud projects describe "$GOOGLE_CLOUD_PROJECT" --
format='value(projectNumber)')
RUNTIME_SA=$(gcloud run services describe suffai-command-center --
format='value(spec.template.spec.serviceAccountName)')
RUNTIME_SA=${RUNTIME_SA:-${PROJECT_NUMBER}-compute@developer.gserviceaccount.com}
TASKS_SA="suffai-tasks-invoker@${GOOGLE_CLOUD_PROJECT}.iam.gserviceaccount.com"

gcloud services enable cloudtasks.googleapis.com
gcloud tasks queues create campaign-rank --location="$GCP_REGION" \
  --max-concurrent-dispatches=10 --max-dispatches-per-second=5 \
  --max-attempts=20 --min-backoff=10s --max-backoff=300s || true
gcloud iam service-accounts create suffai-tasks-invoker \
  --display-name="Cloud Tasks invoker for Command Center" || true

gcloud projects add-iam-policy-binding "$GOOGLE_CLOUD_PROJECT" \
  --member="serviceAccount:$RUNTIME_SA" --role="roles/cloudtasks.enqueuer"
gcloud iam service-accounts add-iam-policy-binding "$TASKS_SA" \
  --member="serviceAccount:$RUNTIME_SA" --role="roles/iam.serviceAccountUser"
gcloud run services add-iam-policy-binding suffai-command-center \
  --member="serviceAccount:$TASKS_SA" --role="roles/run.invoker"

CC_URL=$(gcloud run services describe suffai-command-center --format='value(status.url)')
gcloud run services update suffai-command-center --timeout=900 \
  --update-env-
vars="CLOUD_TASKS_QUEUE=projects/$GOOGLE_CLOUD_PROJECT/locations/$GCP_REGION/queues/campaign-
rank,TASKS_SERVICE_ACCOUNT=$TASKS_SA,COMMAND_CENTER_URL=$CC_URL"
```

> `COMMAND_CENTER_URL` must be the exact service address with no path after it — it is what the job's login token is checked against.

Step 6 — Service account permissions (one-time; harmless to repeat)

```
for role in roles/aiplatform.user roles/bigquery.jobUser roles/bigquery.dataViewer \
            roles/storage.objectViewer roles/cloudsql.client; do
  gcloud projects add-iam-policy-binding "$GOOGLE_CLOUD_PROJECT" \
    --member="serviceAccount:$RUNTIME_SA" --role="$role"
done
# Signed image URLs (Style Assistant, campaign monitor)
gcloud iam service-accounts add-iam-policy-binding "$RUNTIME_SA" \
  --member="serviceAccount:$RUNTIME_SA" --role="roles/iam.serviceAccountTokenCreator"
# Command Center may call the private extraction + similarity services
for s in suffai-extraction suffai-similarity; do
  gcloud run services add-iam-policy-binding $s \
    --member="serviceAccount:$RUNTIME_SA" --role="roles/run.invoker"
done
```

Ranking uploads SKU images to `GCS_IMAGE_BUCKET`, so the runtime account also needs `roles/storage.objectCreator` on that bucket.

Step 7 (optional) — Refresh the chat's metrics snapshot

The dashboard chat reads `web/metrics.json`, which is baked into the image. To refresh it from BigQuery:

```
python3 command_center/scripts/aggregate_metrics.py --bq --project "$GOOGLE_CLOUD_PROJECT"
# then repeat Step 4 so the new file is in the image
```

Step 8 — Smoke test

```
CC_URL=$(gcloud run services describe suffai-command-center --format='value(status.url)')
curl -s "$CC_URL/health" | jq
curl -s -o /dev/null -w '%{http_code}\n' -X POST "$CC_URL/v1/style-assistant/recommend"   #
expect 404/405 (removed)
```

Then click through the main flows:

1. **Brand login → Ads Lab.** Select PINs (try "Select all"), upload 2–3 SKU images, click **Extract & rank**. The job card runs; near the end it shows "All images ranked · saving Style Assistant picks for N customer(s)".
2. **AURA chat on Ads Lab.** After ranking, a "Run on" dropdown (1P / 3P) appears. Choose 1P → one setup form with every field → **Save draft & review** → **Confirm**. The impact and analysis view opens on the Campaigns tab.
3. **Customer login → Style tab** (the only tab a customer sees). Three tabs (Complete your outfit, Recommended, Your wardrobe) and a Brand filter; no Generate button. Picks appear only for customers whose wardrobe PIN is in the ranked campaign.
4. **Admin login.** Admin multi-selects and Geo dropdowns have "Select all". There is no Style Assistant tab.
5. **Access check.** Style Assistant is customer-only: brand and admin logins don't see the tab, and `GET /v1/style-assistant/feed` returns 403 for them.

If something fails

```
gcloud run services logs read suffai-command-center --limit=100
```

Symptom	Likely cause
Job stuck in "queued"	Step 5 settings or permissions missing
Database errors ("column does not exist", status-check violation)	A migration was skipped (Step 2)
Empty Style tab	Campaign ranked before this release (re-rank it), or no wardrobe owners in those PINs
Images missing	Token-creator permission (Step 6)
Login fails	Credentials not loaded into BigQuery (Step 1)

Rollback

Cloud Run keeps earlier revisions:

```
gcloud run revisions list --service=suffai-command-center
gcloud run services update-traffic suffai-command-center --to-revisions=REVISION_NAME=100
```

Notes

- Campaigns ranked before this release have no saved Style picks. Re-run Extract & Rank on any you want customers to see.
- `docs/GCP_REDEPLOY_E2E.md` is out of date (says the prepare script applies migration 005 and lists the removed Style endpoints). Follow this guide instead.